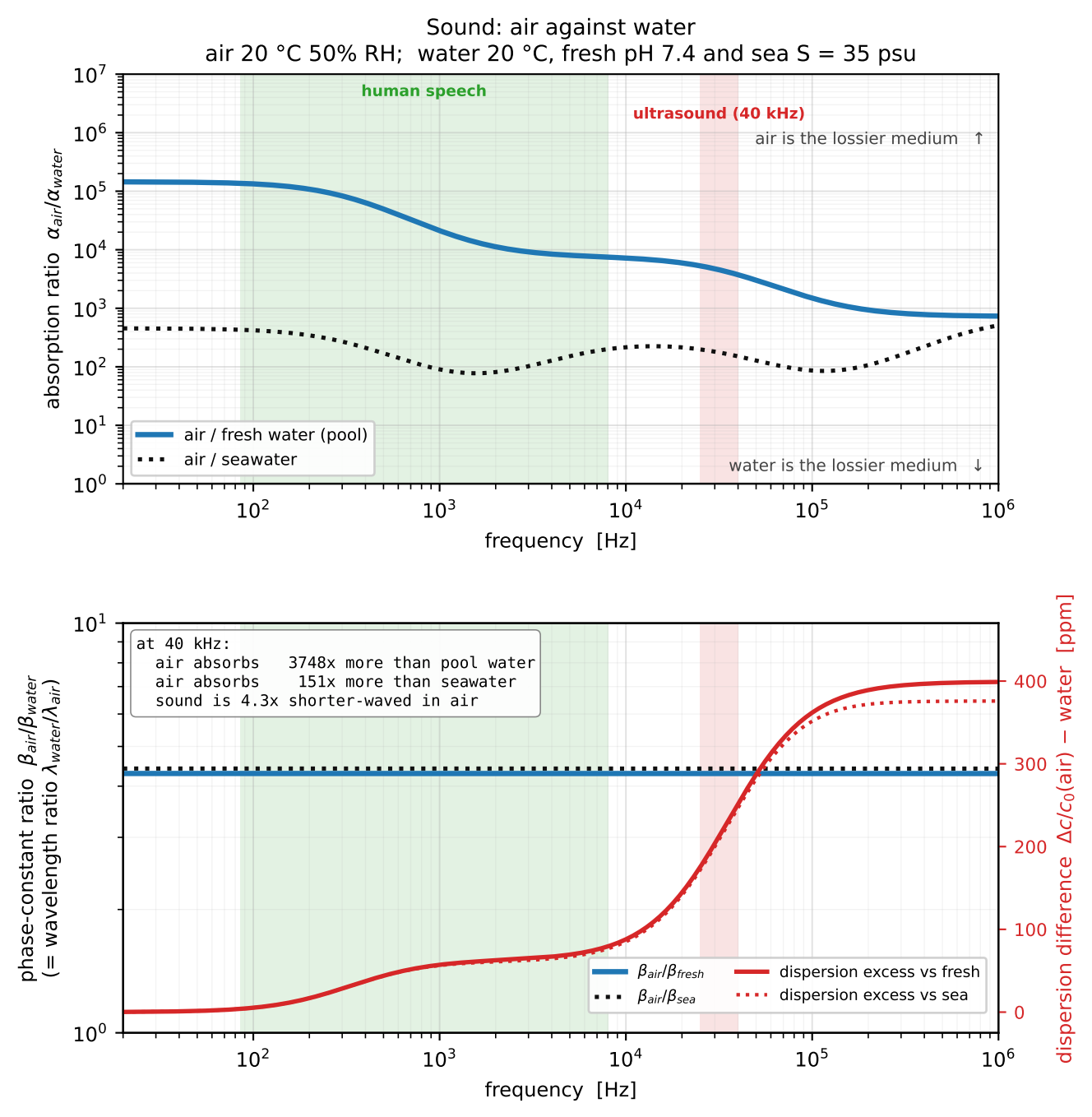
Sound: air against water
air 20 °C 50% RH; water 20 °C, fresh pH 7.4 and sea S = 35 psu
human speech
ultrasound (40 kHz)
air is the lossier medium ↑
water is the lossier medium ↓
absorption ratio $\alpha_{air}/\alpha_{water}$
air / fresh water (pool)
air / seawater
frequency [Hz]
at 40 kHz:
air absorbs 3748x more than pool water
air absorbs 151x more than seawater
sound is 4.3x shorter-waved in air
phase-constant ratio $\beta_{air}/\beta_{water}$
(= wavelength ratio $\lambda_{water}/\lambda_{air}$)
dispersion difference $\Delta c/c_0$(air) − water [ppm]
$\beta_{air}/\beta_{fresh}$
β_{air}/β_{sea}
dispersion excess vs fresh
dispersion excess vs sea
frequency [Hz]